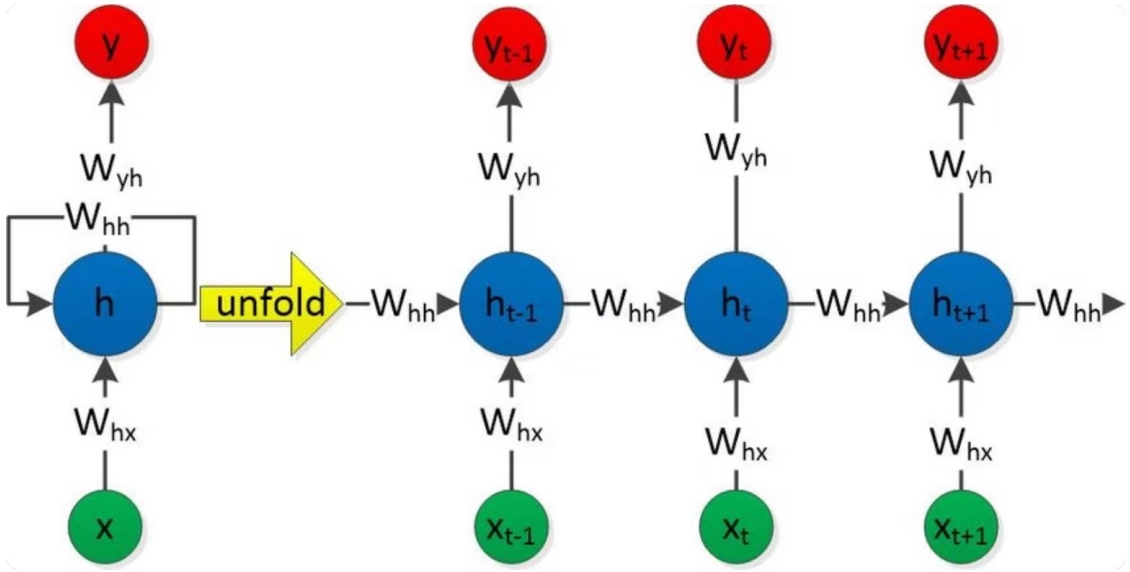
Attention Is All You Need

Almog Schwartzberg
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RNN - Recurrent Neural Network



How RNNs Work?

Hidden state updated at each step:

$$h_t = f(h_{t-1}, x_t)$$

Where x_t = input token, h_{t-1} = previous hidden state, f = learned transformation.

Minimum $O(n)$ sequential operations.

ⓘ "The **apples** in the basket on the table **are** red" — "apples" and "are" are distantly connected, but the signal weakens across many steps.

Why Was a New Model Needed?

Parallelization
(מקביליות)

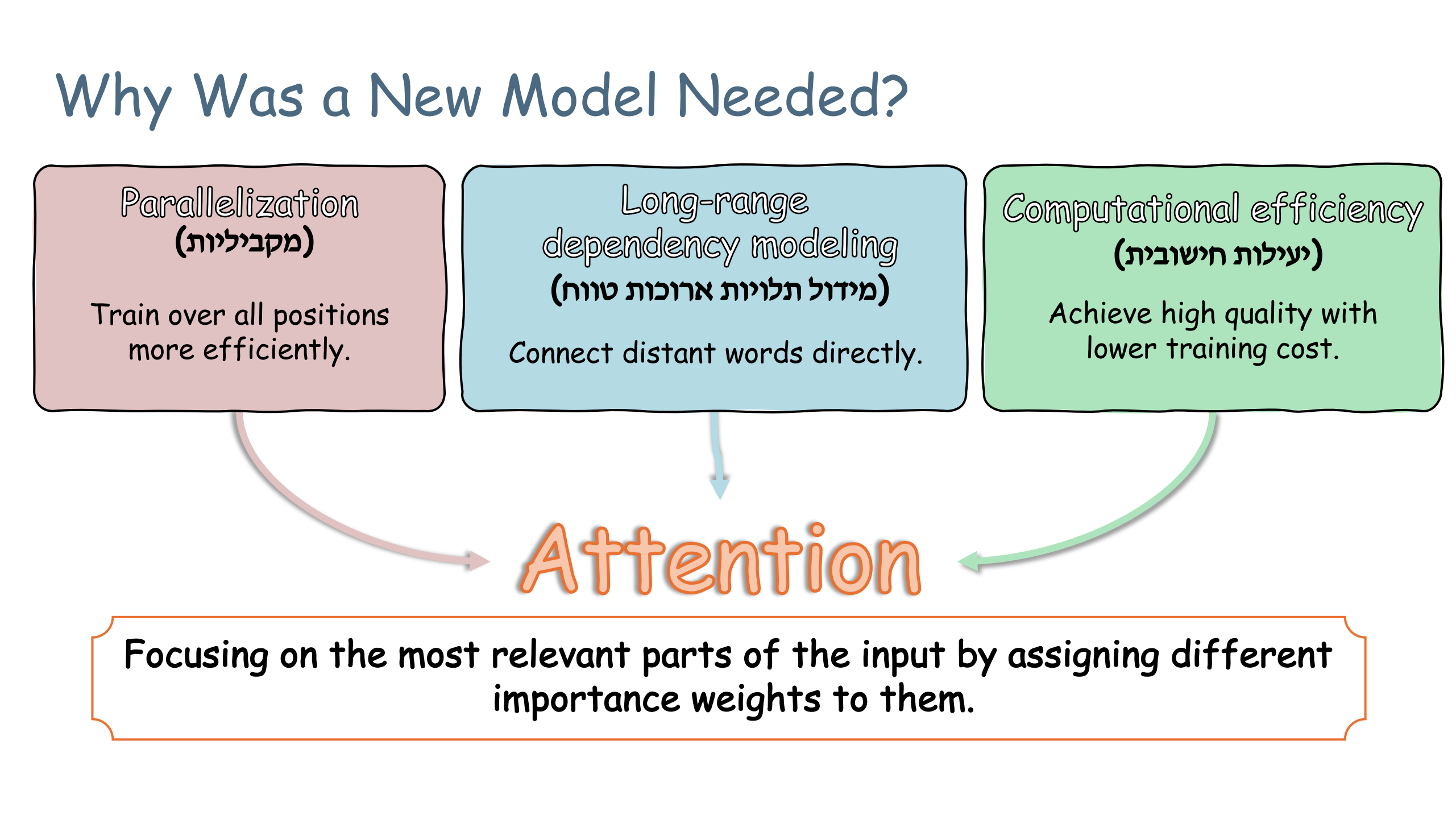
Train over all positions more efficiently.

Long-range dependency modeling
(מידול תלויות ארוכות טווח)

Connect distant words directly.

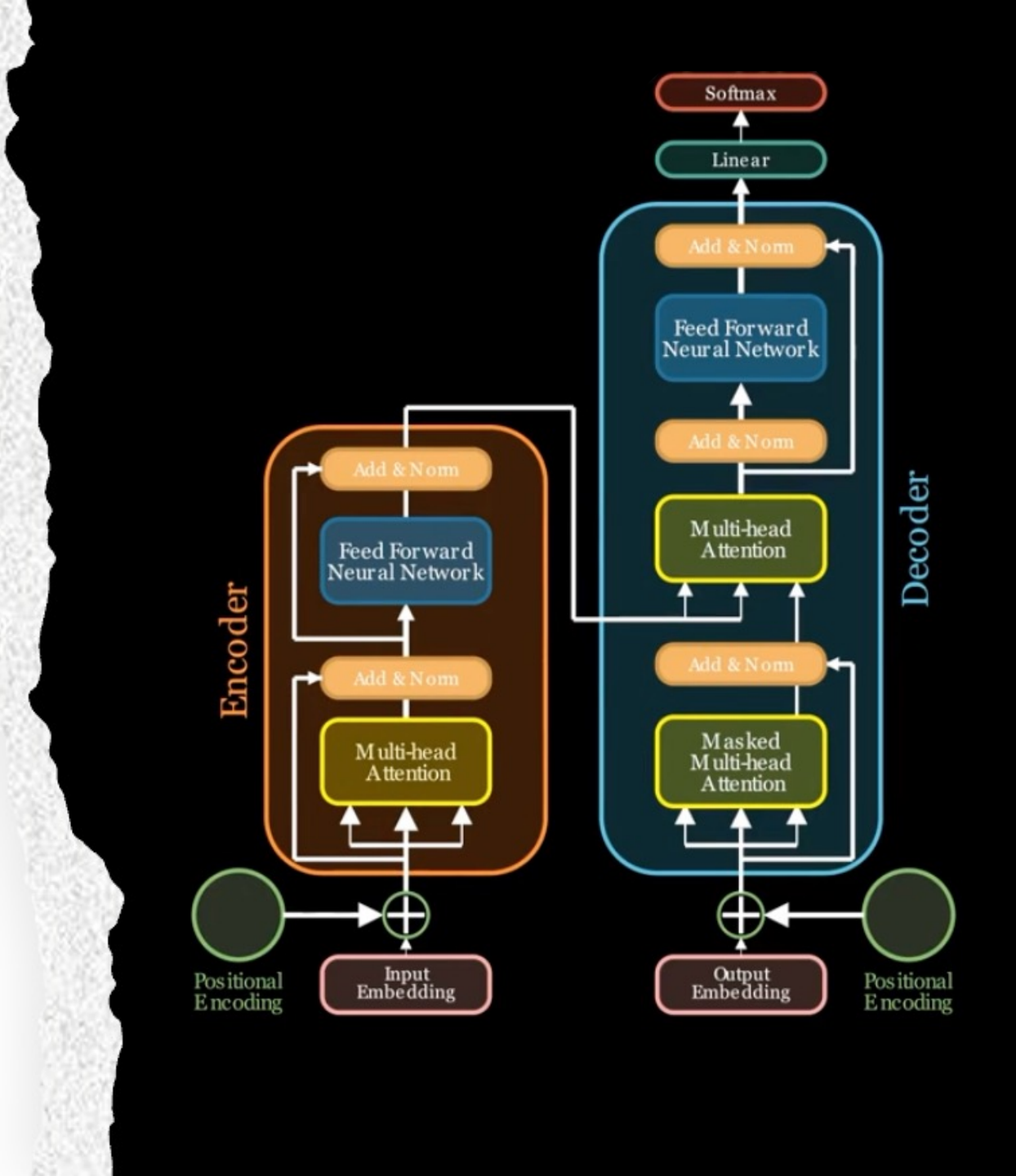
Computational efficiency
(יעילות חישובית)

Achieve high quality with lower training cost.



Focusing on the most relevant parts of the input by assigning different importance weights to them.

The Solution: Transformer



Attention: The Core Mechanism

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{1}{\sqrt{d_k}} \cdot QK^T\right) \cdot V$$

Q (Query)
What I'm looking for.

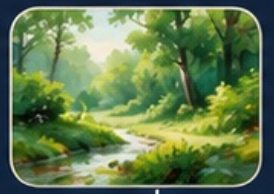
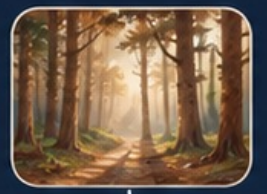
K (Key)
What each token offers.

V (Value)
The token's content

$\sqrt{d_k}$
Scaling factor

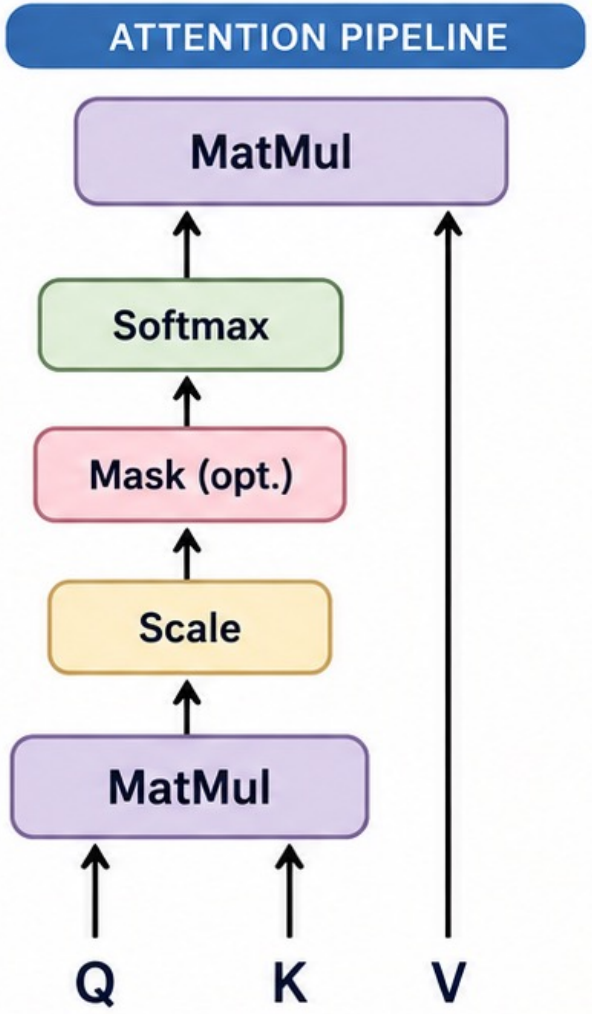
Example: token embeddings



a	fluffy	blue	creature	roamed	the	verdant	forest
$\vec{E}_1$	$\vec{E}_2$	$\vec{E}_3$	$\vec{E}_4$	$\vec{E}_5$	$\vec{E}_6$	$\vec{E}_7$	$\vec{E}_8$

Scaled Dot-Product Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{1}{\sqrt{d_k}} \cdot QK^T\right) \cdot V$$

$$\downarrow$$

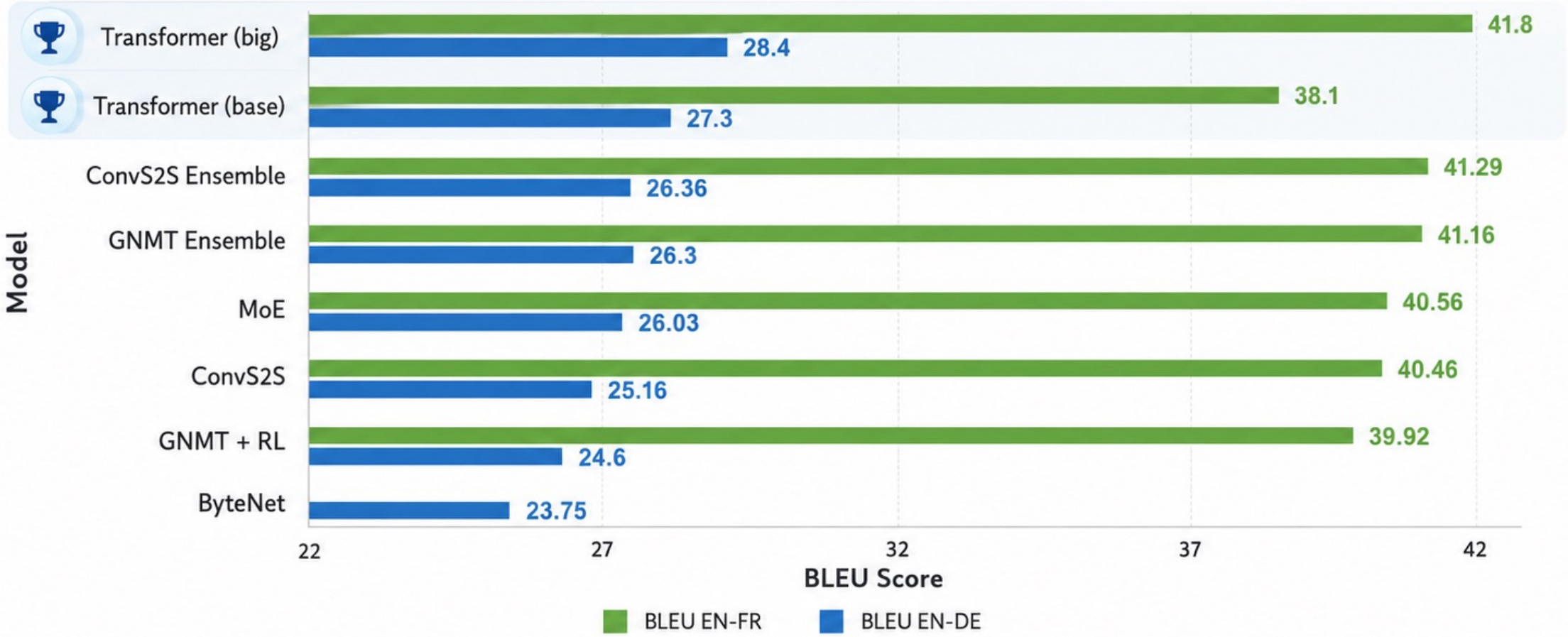
Attention for "it" =

$$= \text{softmax}\left(\frac{1}{\sqrt{d_k}} \cdot Q_{it} \cdot K^T\right) \cdot V =$$

$$= \ldots = \mathbf{0.9} \cdot V_{\text{laptop}} + \mathbf{0.1} \cdot V_{\text{coffee}}$$

Results: BLEU score

We measure the 'accuracy' with BLEU score: built specifically for tasks like machine translation.



Transformer models achieve the strongest BLEU scores, especially on EN-DE, while remaining highly competitive on EN-FR.

Complexity

Model Type	Time Complexity	Sequential Ops	Max Path Length
★ **Self-Attention**	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
RNN	$O(n \cdot d^2)$	$O(n)$	$O(n)$
CNN	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k n)$

- $\boldsymbol{n}$ = sequence length (number of tokens in the input)
- $\boldsymbol{d}$ = representation dimension (size of vector embeddings)
- $\boldsymbol{k}$ = convolution kernel size (for CNNs)

Self-attention demonstrates particular efficiency under certain conditions.
It is most efficient when $n < d$.

This comparison highlights that while self-attention has a quadratic dependency on sequence length (*n^2*), it excels in having constant sequential operations and maximum path length. This makes it highly parallelizable, a critical advantage for training on modern hardware, especially for tasks like sentence-level translation where *n* is often smaller than *d*.

What Comes After Transformer?

The original "Attention Is All You Need" paper didn't just introduce a groundbreaking architecture; it also laid the groundwork for numerous future research avenues. Many of the challenges and opportunities hinted at in 2017 have become central to modern AI development.

- Do language models truly understand, or do they only predict?
- Will the Transformer remain the dominant architecture for AI?
- Is full attention sustainable and efficient for very long sequences?
- How should we balance performance, cost, and ethical responsibility in AI development?
- What might be the next major breakthrough after attention?

Open questions for the future of AI